

Lab 3: Component Modelling & Architectural Pattern Selection

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System: Self-Service Coffee Kiosk

Architecture Selection

We chose **Layered Architecture** for the Self-Service Coffee Kiosk System. The system is divided into Presentation, Business, and Data layers, with each layer handling a specific part of the kiosk functionality.

Reason 1 – Separation of Responsibilities

The Presentation Layer contains the User Interface component, which handles touchscreen interaction. The Business Layer contains the Order Manager, Payment Service, and Receipt Printer components, allowing order processing, payment handling, and receipt printing to remain separate. This makes the system easier to understand and maintain.

Reason 2 – Clear Data Management

The Data Layer contains the Database component for storing menu and pricing information. The Order Manager communicates with the Database through a defined interface instead of directly mixing data storage responsibilities with the user interface or payment functions.

Security Advantage

Layered Architecture provides controlled access between parts of the system. The User Interface does not directly access the Database; requests are handled through the appropriate business component and defined interfaces. This helps reduce unnecessary direct access to stored data.

Performance Benefit

Separating the system into focused components allows individual operations such as order processing, payment handling, receipt printing, and database access to be handled independently. This can reduce unnecessary interaction between unrelated parts of the kiosk and makes individual components easier to optimize.

Components

1. User Interface 2. Order Manager 3. Payment Service 4. Receipt Printer 5. Database

Interfaces

1. Order Interface 2. Payment Interface 3. Receipt Interface 4. Database Interface